

Introduction to Microeconomics

Lecture 1—Logistics for the course & Overview of Economics and Research in Economics

Prof. Alejandro Melo-Ponce



Years
Anniversary

ECON101
Spring 2026

Who I am, where I am, and how can you find me?

- Office 8.510, however, sometimes I prefer to meet students in one of the coffee places around the University.
- Office hours by apointment, book slots by [youcanbookme](#): the link is in the syllabus. Every week I open some slots which you can book.

- Disregard the Registrar's Tutorial sections!
- I coordinate new times with the help of my TAs.
- Announcements about this will be made by the end of this week.

There is no official textbook for the course. However, the following references might be useful:

- Shapiro, David. *Principles of Macroeconomics* 3e. OpenStax College, 2022.
- Börgers, Tilman. *Lecture Notes on Microeconomic Theory*.
- Perloff, Jeffrey M. *Microeconomics*. Pearson, 2018, 8th ed.
- Frank, Robert; Bernanke, Ben; Antonovics, Kate; Heffetz, Ori. *Principles of Microeconomics*. McGraw Hill 8th ed.

Activity	Date	Weighting
Midterm Exam 1	Week 6	30%
Midterm Exam 2	Week 10	30%
Final	TBA	40%

A	95–100	A–	90–94.99
B+	85–89.99	B	80–84.99
B–	75–79.99	C+	70–74.99
C	65–69.99	C–	60–64.99
D+	55–59.99	D	50–54.99
F	0–44.99		

Why not have more ECON 101 sections?

Should there be 6 sections of ECON101 or only 1?

Students learn more effectively in smaller classes.

But smaller classes are also more expensive.

Faculty salary + Usage of classroom (electricity, heating, IT support, etc.): \$ 45,000 USD.

Per student cost :

- One section $\$45,000/150 = \300 .
- Six sections (25 students per section) $\$45,000/25 = \1800 .

Would society (the tax payers) be willing to pay the extra \$1500 to switch from one section to six?

The problem

- Children are supposed to be picked up by 4:00 PM.
- Some parents are late which leads to →
 - Anxious children.
 - At least one teacher that has to wait until all parents arrive.

The proposal

- Two economists suggested to fine tardy parents.
- Any parent arriving more than 10 minutes late would pay \$ 3 per child for each incident. The fee would be added to the monthly bill (avg. \$380)

The result

- Before the fine: avg. 8 late pickups per week per day-care center.
- After the fine: avg. 20 late pickups per week. . . Why?

Incentives!

Economic Models

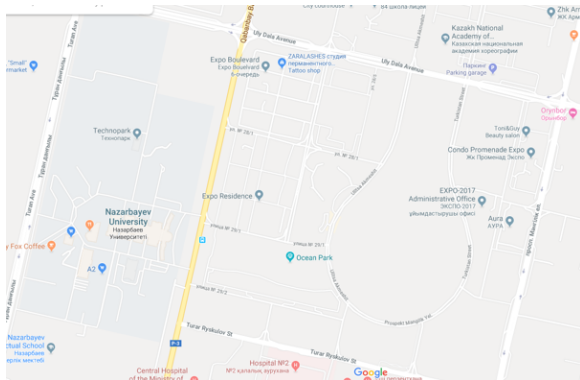
Models are simplified representations of the real world used as the basis for predictions or explanations

Assumptions

Assumptions are explicit restrictions on behavior, information, technology, and institutions that define how the model works.

Crucial implication: The collection of assumptions determines the *domain of applicability* of the model.

Examples of Models



- **Ceteris paribus** means: *all else equal*.
- When studying the effect of one factor, we assume that all other relevant factors remain unchanged.
- This allows us to perform a controlled *thought experiment*.

Why it matters

A good economic model allows us to ask:

What would happen if we changed only one thing?

- Observing that two variables move together does *not* imply that one causes the other.
- Without holding other factors fixed, we can only observe **correlations**.
- Ceteris paribus reasoning is what allows us to think about **causal effects**.

A study in England found that:

Cows that were given names produced more milk than cows without names.

Is this:

- Causation?
- Or merely correlation?

What could be going on?

Why Correlation Is Not Causation

Two common problems:

- ① **Omitted variables:** A missing factor affects both variables, creating a misleading correlation.
- ② **Reverse causality:** The direction of cause and effect runs opposite to what we initially think.

Key takeaway

Economic models, through their assumptions and ceteris paribus reasoning, help us distinguish correlation from causation.

Causation: One factor directly affects another.

Correlation: Two variables move together, but not necessarily because one causes the other.

- Correlation can be positive or negative.
- Correlation alone is not enough for policy or prediction.

The Scarcity Principle (also called the There-is-no-free-lunch principle)

Although we have unlimited wants and needs, the resources available to us are limited. So having more of one good thing usually means having less of another.

This is the ♥ of the Economic problem!→ This means there are trade-offs: if you want more of one thing you need to give up another.

Microeconomics

- Microeconomics entails the study of how economic agents choose under scarcity.
- Institutions (markets, rules, customs, etc.) create **incentives**.
- **Incentives** alter the trade-offs economic agents face → Microeconomics is the study of **incentives and how economic agents respond to them!**

The Rationality assumption

People have preferences/goals, and try to fulfill them as best as they can.

Cost-Benefit analysis

Should I do activity x ?

- $C(x)$ = the cost of doing x .
- $B(x)$ = the benefits of doing x .
- If $B(x) > C(x)$, do x , otherwise, do not.

Can cost-benefit analysis help you make better decisions?

Should you get up to change the turntable?

You are a music connoisseur and you have settled into a comfortable chair after putting a vinyl album in the turntable. Then you realize that the next two tracks on the album are ones you hate. If instead of listening to the album in vinyl, you used Spotify, you could have programmed a playlist omitting the tracks you dislike. But you didn't, so now you must decide whether to get up and skip the tracks manually, or stay put and wait it out.

- Suppose that $C(x) = \$1$, which is the minimum amount it would take to get you out of your chair.
- $B(x)$ is the maximum you would be willing to pay someone to turn down the volume.

If $B(x) > \$1$, then get up to skip the tracks. If $B(x) < \$1$, then don't.

- You are about to buy a 10,000 KZT alarm clock at the Technodom in Mega Silkway, in front of the NU campus. However, one of your friends tells you that the same alarm clock is on sale at the Republic Avenue Technodom, on the right bank, for 5000 KZT.
- You are about to buy a 500,000 KZT laptop at the Technodom in Mega Silkway. However, one of your friends tells you that the same laptop is on sale at the Republic Avenue Technodom, for 495,000 KZT.

What do you do?

The framing of the choice sometimes matters. . .

The Coronavirus will claim 600 lives if we do nothing. We must choose between these two programs:

- Program A: Save 200 lives with certainty.
- Program B: Save 600 lives with probability $1/3$; save zero lives with probability $2/3$.

Same Coronavirus, now consider

- Program C: 400 people will die with certainty.
- Program D: $1/3$ chance that no one will die; $2/3$ chance that all 600 will die.

Losing a 20,000 KZT bill

You have just arrived at the theater to buy your ticket and discover that you have lost a 20,000 KZT bill from your wallet. Do you still buy the ticket and see the play anyway?

A lost theater ticket

You have just arrived at the theater and discover that you have lost the 20,000 KZT ticket that you purchased through Ticketon earlier that day. Do you buy another ticket and see the play anyway?

How much memory should your computer have.

Suppose that RAM can be added to your computer at \$15 per gigabyte. Suppose also that the value as measured in terms of your willingness to pay, of an additional gigabyte of memory, is as shown on the graph below.

Definition: Opportunity Cost

The opportunity cost of activity X is the total surplus (value) of the next-best alternative Y that you did not take.

Which concert should you go?

You won a free ticket to see BTS. The ticket has no resale value.

BlackPink is performing on the same night and is your next-best alternative.

Tickets to see BlackPink cost \$40.

You would be willing to pay up to \$50 to see BlackPink.

Assume there are no other costs of seeing either group. What is the opportunity cost of attending the BTS concert?

- ① 0
- ② 10
- ③ 40
- ④ 50

Variation 1

Now suppose that you are willing to pay as much as \$70 to see BlackPink.

The cost of a BlackPink ticket is \$30.

What is the opportunity cost of seeing BTS?

Variation 2

Same setup as before, but now you must buy the BTS ticket, which costs \$60. You are willing to pay as much as \$70 to see BlackPink.

The cost of a BlackPink ticket is \$30.

What is the opportunity cost of seeing BTS?

Variation 3

Same setup as variation 2, but now we ask, what is the minimum amount you'd have to be willing to pay to see BTS to make their concert a better choice?

Variation 4

Same as before, except now you have bought a **nonrefundable** \$60 ticket to see BTS.

Some Common Pitfalls for Decision Makers

Pitfall #1: Measuring Costs and Benefits as proportions rather than as absolute money amounts (like the Technodom example)

Pitfall #2: Ignoring Implicit Costs

If doing activity x means not being able to do activity y , then the value of you doing y is an **implicit cost** of doing x . You will end up doing bad decisions (and not behaving like a true economist!) if you forget the value of forgone opportunities. Thus, questions like

“Should I do x ?

should always be translated into ones like

Should I do x instead of y ?

where y is the most highly valued alternative to doing x .

Should you go skiing today?

In an ideal world, the NU campus should be in Almaty, and preferably, close to Shymbulak. Suppose we are in this ideal world. From experience you can say that a day on the slopes is worth \$50 to you. The charge for the day is \$30 (transportation and lift ticket).

If you don't go skiing, you will work at your new job as research assistant for Prof. Alejandro. The job pays \$40 per day, and you like it so much that you would have been willing to do it for free.

Should you go skiing or stay and work as research assistant?

Do psychic cost of doing the job matter?

Suppose now that instead of being a research assistant, which you love, you would instead have a job scraping plates at the NU Canteen, which pays the same \$40 a day. However, this job is so unpleasant to you that you would be unwilling to do it for less than \$25 a day. Assume your manager at the canteen permits you to take the day off whenever you want. What should you do?

Is it fair to charge interest when lending a friend some money?

Suppose a friend lends you \$10,000, and her primary concern when deciding whether to charge interest is to decide if it is fair to do so. She could have put that same money in the bank, where it would have earned, say, 5% interest, or \$500 each year. If she charges you \$500 interest for each year the loan is out, she is merely recovering the opportunity cost of her money. If she didn't charge you any interest, it would be the same as making you a gift of \$500/yr.

Why do banks pay interest for savings deposits?

Suppose you owned a bank and someone deposited \$10,000 in it on January 1 without your having to pay him interest. You could then take the money and buy a productive asset, such as a stand of trees. Suppose that each year trees grow at the rate of 6%, and that the price of a tree is proportional to the amount of lumber in it. At the end of the year you could then sell the trees for \$10,600, and have \$600 more than before. But that same option was available to the person who put his money in your bank. Why should he give you the \$600 he could have earned? He will be willing to let you use his money only if you compensate him for the opportunity cost of not using it himself. If you pay him 5% interest, he will get \$500, which will probably be acceptable to him because he won't have to go to the trouble of tending the trees himself (or of lending the money to someone who will tend them). You get to keep the remaining \$100 for taking care of that.

Failure to ignore sunk costs.

An opportunity cost will often not seem like a relevant cost when in reality it is. Another pitfall in decision making is that sometimes an expenditure will seem like a relevant cost when in reality it is not. Such is often the case with **sunk costs**, costs that are beyond recovery at the moment a decision is made. Unlike opportunity costs, these costs should be ignored

Should I drive to Pavlodar?

You are planning a 400 KM trip to Pavlodar (from Astana). Except for the cost, you are indifferent between taking the bus or driving. Bus fare is \$100. You don't know how much it would cost to drive your car, so you call the local rental agency for an estimate. The attendant tells you that for your make of car the costs of a typical 16,000 KM driving year are as follows:

Insurance	\$1000
-----------	--------

Interest	\$2000
----------	--------

Fuel and oil	\$1000
--------------	--------

 These costs come to \$0.3/kilometer, and you use this figure to

Maintenance	\$1000
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Total	\$5000
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compute that the 400 KM trip to Pavlodar will cost you \$120. Since this is more than the bus fare, you decide to take the bus.

Was the previous calculation correct?

No! Because you committed the error of not ignoring sunk costs! The insurance and interest payments do not vary with the number of kilometers you drive each year. Both are **sunk costs** and will be the same whether you drive to Pavlodar or not. Of the costs listed, only fuel & oil and maintenance are the only ones that vary with the miles driven. These total \$2000, or if you drive 16,000 kilometers yearly, \$0.125/kilometer. Thus, it would cost you to drive to Pavlodar \$50, since this is much less than the bus fare, you should drive.

Pitfall #4: Failure to understand the average-marginal distinction

Suppose you own a fishing fleet consisting of a given number of boats, and can send your boats in whatever numbers you wish to either of two ends of an extremely wide lake, east or west. Under your current allocation of boats, the ones fishing at the east end return daily with 100 kilograms of fish each, while those in the west return daily with 120 kilograms each. The fish populations at each end of the lake are completely independent, and your current yields can be sustained indefinitely. True or False: If you shift some of your boats from the east end to the west end, you will catch more fish?

Fishing example: more data

In the fishing fleet scenario just described, suppose the relationship between the number of boats sent to each end and the number of pounds caught per boat is as summarized in the table. Suppose further that you have four boats in your fleet, and that two currently fish the east end while the other two fish the west end. Should you move one of your boats from the east end to the west end?

Number of boats	East end	West end
1	100 kg/boat	130 kg/boat
2	100 kg/boat	120 kg/boat
3	100 kg/boat	110 kg/boat
4	100 kg/boat	100 kg/boat

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Current output (2 east, 2 west)=440 kg.

Output with 1 east, 3 west=430 kg.

How should you allocate resources?

The general rule for allocating a resource efficiently across different production activities is to allocate each unit of the resource to the productive activity where its marginal benefit is highest.

For a resource that is perfectly divisible, and for activities for which the marginal product of the resource is not always higher in one than in the others, the rule is: allocate the resource so that its marginal benefit is the same in every activity.

Similar fishing example

Number of boats	East end	West end
1	100 kg/boat	120 kg/boat
2	100 kg/boat	120 kg/boat
3	100 kg/boat	120 kg/boat
4	100 kg/boat	120 kg/boat

Seven important principles

- ① The scarcity principle
- ② The cost-benefit principle
- ③ The incentive principle
- ④ The principle of comparative advantage.
- ⑤ The principle of increasing opportunity cost.
- ⑥ The equilibrium principle.
- ⑦ The efficiency principle.



What topics does Economics study?

- A large variety of topics!
- Almost all domains of human activity can be studied with Economics!
- Economics research is organized into fields.
- The goal today is to imprint on you the role that our introductory course in microeconomic theory relates to the universe of contemporary economics research.

- Let's focus on the article by Antràs and coauthors.
- Focuses on firm's choices of which inputs for their production to import from abroad.
- Why do you think this is relevant?
- What do they find?

- The conventional structure for a research article is to have an introduction, and at the end of this introduction, there is typically a paragraph that explains how the paper is structured.
- Quoting from the article:

*The rest of the paper is structured as follows. We present the assumptions of our **model** in Section I, and solve for the equilibrium in Section II. In Section III we introduce the **data** and provide descriptive evidence supporting the assumptions underlying our theoretical framework. We estimate the model structurally in Section IV, and in Section V, we perform our counterfactual analysis and compare the predictions of the model to reduced-form evidence. Section VI concludes.*

- The data comes from US Customs and Borders Protection.
- It also includes data from other US government agencies.
- Some of it is confidential.

Consider a world consisting of J countries in which individuals value the consumption of differentiated varieties of manufactured goods according to a standard symmetric CES aggregator

$$U_{M_i} = \left(\int_{\omega \in \Omega_i} q_i(\omega_i)^{(\sigma-1)/\sigma} d\omega \right)^{\sigma/(\sigma-1)} \quad (1)$$

- A **model** is something mathematical.
- A mathematical description of how that aspect of the economy works:
 - How inputs are transformed into outputs, how decisions about inputs and outputs are made, how much of the output firms will be able to sell, how this depends on the prices firms choose, how firms will set their prices, etc.
 - Specifically, for equation (1), it describes how much enjoyment people derive in the model from consumption of manufactured goods.
 - The LHS is a symbol for “enjoyment” (**utility** in the jargon of Economics).
 - The RHS is a formula that involves quantities of the various goods a person buys.
 - In the model, the utility will determine the demand for manufactured goods.
 - The demand, in turn, contributes to the decision that firms make about which inputs to buy from abroad.
- But, how can a mathematical description of manufactured goods, imports, prices, come **before we see the data?**

- Let's take another look:

$$U_{M_i} = \left(\int_{\omega \in \Omega_i} q_i(\omega_i)^{(\sigma-1)/\sigma} d\omega \right)^{\sigma/(\sigma-1)}$$

- What is the value of σ ?
- That is what the authors want to figure from the data!
- After invoking their data, they estimate the value to be $\sigma = 3.85$.

- Before even seeing the data, the authors make assumptions about the consumers' utility, as expressed by (1).
- Then, they look at the data assuming that this formula is correct, leaving the only remaining question to be what is the value of σ .
- There are many other assumptions: how firms transform inputs into outputs, the market structure, etc.
- The model is just a **lens or filter** through which the authors look at their data, and this lens, together with what is out there, determine what the authors focus on.

Why to use a model then?

- Don't lenses distort what you see?
- Models allow economists to focus on a particular aspect of what they see.
- In this example, it permits the authors to focus on what they see in their data, and prevents them from getting lost in the sea of data.
- It allows them to **focus and what they want to investigate**.
- Another benefit: a model allows researchers to ask questions about economic policy: if the model and derivations are correct, we can test new government policies that don't yet exist within the model to simulate their effects.

Does the model fit perfectly the data?

- Of course not, that would be miraculous.
- There are random effects not captured by the model.
- To cope with these, the authors use methods from statistics.
- The specific methods that we have developed to analyze economic data is called **econometrics**.

- ① Find an interesting question.
- ② Develop a mathematical model that is suitable.
- ③ Use data to figure out the value of the parameters.
- ④ Use the understanding of the world developed in this exercise to make predictions.

Do all economic research look like this?



What is the role of Microeconomic Theory

- Two skills that you will develop as an Economist:
 - The ability to write suitable mathematical models.
 - The econometric skills needed when taking models to real world data.
- Two fields concerned with developing the tools and ideas used in these two skills:
 - Theory
 - Econometric Theory
- Research in these fields often looks like pure mathematics rather than what a layperson would consider “economics”.
- This class is an introduction to the first set of skills:
 - You will be introduced to simplified versions of the main tools that appear in Economic Models.
 - Think of Economic Models as a **fable**: in this class you will meet the simple version of the characters that populate economic fables.
 - The models might look too simple or naive, you might feel that they are too disconnected from the real world. But even in this setting you will soon encounter things that look similar to (1) and hopefully you will soon understand how such things are used in Economics.

- Macroeconomics is the study of fluctuations in the total value of production in an economy, and with the growth and shrinkage of that production.
- Is there a difference between macroeconomic and microeconomic theory?
- The distinction is old fashioned (1960's).
- Modern macroeconomic theory¹ uses tools from microeconomic theory.

¹(That is, good macroeconomic theory! Caveat: this is a personal opinion, don't tell your local macroeconomists unless you have more information about their preferences!)